

ASSIGNMENT III

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write a function to compute.

- CGPA, Grading
- pearson
- Kendal
- spear man and put them in data frame.

```
GPA <- function()
{ # creating an empty dataframe with three user input variables
  scores <- data.frame(
    Course_Code = character(),
    marks = numeric(),
    credit_Hours = numeric()
  )
  # editing the dataframe by keying in user values
  score <- edit(scores)

  # Defining some vectors
  mrks <- as.vector(score$marks)
  credit <- as.vector(score$credit_Hours)
  Grade <- c()
  GPT <- c()

  for(i in 1:length(mrks))
  {
    if (length(score$Course_Code) != length(score$marks) |
length(score$Course_Code) != length(score$credit_Hours))
    {
      print("Please make sure all the variables are of the same length")
      break
    }
  }
}
```

```

}
else
  if(mrks[i]> 100)
  {
    grade <- 'N/A'
    gpt <- 0*(credit[i])
  }
  else if(mrks[i] >= 80)
  {
    grade <- "A"
    gpt <- 4.0*(credit[i])
  }
  else if(mrks[i] >= 75)
  {
    grade <- "B+"
    gpt <- 3.5*(credit[i])
  }else if(mrks[i] >= 70)
  {
    grade <- "B"
    gpt <- 3.0*(credit[i])
  }else if(mrks[i] >= 65)
  {
    grade <- "C+"
    gpt <- 2.5*(credit[i])
  }else if(mrks[i] >= 60)
  {
    grade <- "C"
    gpt <- 2.0*(credit[i])
  }else if(mrks[i] >= 55)
  {
    grade <- "D+"
    gpt <- 1.5*(credit[i])
  }else if(mrks[i] >= 50)
  {
    grade <- "D"
    gpt <- 1.0*(credit[i])
  }else if(mrks[i] >= 45)
  {
    grade <- "E"
    gpt <- 0.5*(credit[i])
  }else if(mrks[i] >= 0)
  {
    grade <- "F"
    gpt <- 0.0*(credit[i])
  }
  else if(mrks[i]<0){
    grade <- 'N/A'
    gpt <- 0*(credit[i])
  }
}

```

```

    Grade[i] <- grade
    GPT[i] <- gpt
  }

  ### calculation for the CGPA
  CCT <- sum(credit) # cumulative credit taking
  CGPT <- sum(GPT) # Cumulative grade point taken

  gpa <- round(CGPT/CCT,2)
  GPA <- c(gpa, rep(x=' ', times = length(mrks)-1))
  result <- data.frame('Course Code' = score$Course_Code, 'Marks (%)' =
mrks, 'Credit Hours' = credit, 'Grade' = Grade, 'Grade Point' = GPT, 'GPA' =
GPA)

  return(result)
}

```

GPA()

```

##   Course.Code Marks.... Credit.Hours Grade Grade.Point GPA
## 1   ACTU 301      98           3    A      12.0 3.5
## 2   ACTU 359      76           3   B+      10.5
## 3   FINC 301      88           3    A      12.0
## 4   MATH 355      65           3   C+       7.5
## 5   STAT 331      86           3    A      12.0
## 6   STAT 333      67           3   C+       7.5
## 7   STAT 339      90           3    A      12.0

```

```

my_correlation <- function(X,Y)
{
  if(length(X) != length(Y))
  {
    print(" X and Y must be of the samme length")
  }
  else
  {
    N <- length(X)
    n = N
    r <- (N*sum(X*Y)-sum(X)*sum(Y))/(sqrt((N*sum(X^2)-
(sum(X))^2)*(N*sum(Y^2)-(sum(Y))^2)))

```

```

    concordant <- c()
    discordant <- c()
    for(i in 1:length(X))
    {
      if(X[i] %in% Y)
      {
        conc <- X[i]

```

```

        concordant[i] <- conc
      }
    else
    {
      disc <- X[i]
      discordant[i] <- disc
    }
  }

  concordant <- discordant <- 0

  for (i in 1:(n - 1)) {
    for (j in (i + 1):n) {
      if ((X[i] - X[j]) * (Y[i] - Y[j]) > 0) {
        concordant <- concordant + 1
      } else if ((X[i] - X[j]) * (Y[i] - Y[j]) < 0) {
        discordant <- discordant + 1
      }
    }
  }

  k <- (concordant - discordant) / choose(n, 2)

  di <- X-Y
  rho <- 1 - ((6*sum(di^2))/(n*(n^2-1)))

}

return(data.frame('Pearson'=round(r,4),
                  'Kendall' = round(k,4),
                  'Spearman' = round(rho,4)))
}

X <- 1:10
Y <- 10:1
my_correlation(X,Y)

##   Pearson Kendall Spearman
## 1      -1      -1      -1

```

Go into the ExpQQ and ensure that

1. No tittle of plot
2. pareto is a special distribution for modelling large data
3. use the ExpQQ to achieve the pareto.

NOTE: If Z has the basic pareto distribution with shape parameter a , then $T = \ln(z)$ has the exponential distribution with rate parameter a .

```
library(ReIns)
#trace(ExpQQ, edit = TRUE)

my_pareto <- function (data, plot = TRUE, main = NULL)
{
  X <- as.numeric(sort(data))
  n <- length(X)
  i <- 1:n
  eqq.the <- ((n+1)/(n+1-i))
  eqq.emp <- X
  plot(eqq.the, eqq.emp, type = "p", xlab = "Quantiles of standard pareto",
       ylab = "X", main = main)
}

library(MASS)
my_pareto(anorexia$Prewt)
```

